

RAJARATA UNIVERSITY OF SRI LANKA, MIHINTALE
FACULTY OF APPLIED SCIENCES
DEPARTMENT OF PHYSICAL SCIENCES

FDN 1204/ FDN 1306 – Basic Mathematics
Tutorial

1. ABC is any triangle and D, E, F are the middle points of sides BC, CA, AB respectively; express
 - i. The vectors $\overrightarrow{BC}, \overrightarrow{AD}, \overrightarrow{BE}$ and $\overrightarrow{CF}$ as linear combination of vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$.
 - ii. The vectors $\overrightarrow{AB}, \overrightarrow{BC}, \overrightarrow{CA}, \overrightarrow{AD}$ as linear combination of vectors $\overrightarrow{BE}$ and $\overrightarrow{CF}$.
 - iii. The vectors $\overrightarrow{AC}, \overrightarrow{BC}, \overrightarrow{AD}$ and $\overrightarrow{CF}$ as linear combination of vectors $\overrightarrow{AB}$ and $\overrightarrow{BE}$. Show that $\overrightarrow{AD} + \overrightarrow{BE} + \overrightarrow{CF} = 0$.

2. ABCD is a parallelogram and AC, BD are its diagonals. Express
 - i. $\overrightarrow{AC}$ and $\overrightarrow{BD}$ in terms of $\overrightarrow{AB}$ and $\overrightarrow{AD}$.
 - ii. $\overrightarrow{AB}$ and $\overrightarrow{AD}$ in terms of $\overrightarrow{AC}$ and $\overrightarrow{BD}$.
 - iii. $\overrightarrow{AB}$ and $\overrightarrow{AC}$ in terms of $\overrightarrow{AD}$ and $\overrightarrow{BD}$.Show that
$$\overrightarrow{AC} + \overrightarrow{DB} = 2 \overrightarrow{DC}, \overrightarrow{AC} - \overrightarrow{BD} = 2 \overrightarrow{AB}$$

3. OABC is a tetrahedron express
 - i. The vectors $\overrightarrow{BC}, \overrightarrow{CA}$, and $\overrightarrow{AB}$ in terms of the vectors $\overrightarrow{OA}, \overrightarrow{OB}$, and $\overrightarrow{OC}$.
 - ii. The vectors $\overrightarrow{OA}, \overrightarrow{OB}$, and $\overrightarrow{CA}$ in terms of the vectors $\overrightarrow{OC}, \overrightarrow{AB}$, and $\overrightarrow{BC}$.